In each of Problems 11 through 28, find the general solution of the given differential equation.

4.2. 12. $y''' - 3y'' + 3y' - y = 0$

3rd order, lin, ODE, homogeneous w/ const. coefs.

char eqn: $r^3 - 3r^2 + 3r - 1 = 0$

$(r-1)^3 = 0 \longrightarrow r_{1,2,3} = 1$ → $y_1 = e^t$, $y_2 = te^t$, $y_3 = t^2e^t$

gen. soln: $y = c_1e^t + c_2te^t + c_3t^2e^t$

In each of Problems 29 through 36, find the solution of the given initial value problem, and plot its graph. How does the solution behave as $t \to \infty$?

4.2. 31. $y^{(4)} - 4y''' + 4y'' = 0$; $y(1) = -1$, $y'(1) = 2$, $y''(1) = 0$, $y'''(1) = 0$

char eqn. $r^4 - 4r^3 + 4r^2 = 0$

$r^2(r-2)^2 = 0$

$r_{1,2} = 0$ → $y_1 = 1$, $y_2 = t$

$r_{3,4} = 2$ → $y_3 = e^{2t}$, $y_4 = te^{2t}$

gen. soln. $y = c_1 + c_2t + c_3e^{2t} + c_4te^{2t}$

$y' = c_2 + 2c_3e^{2t} + c_4e^{2t} + 2c_4te^{2t}$

$y'' = 4c_3e^{2t} + 2c_4e^{2t} + 2c_4e^{2t} + 4c_4te^{2t}$

$y''' = 8c_3e^{2t} + 4c_4e^{2t} + 4c_4e^{2t} + 4c_4e^{2t} + 8c_4te^{2t}$

IVC $y(1) = -1 \rightarrow c_1 + c_2 + c_3e^2 + c_4e^2 = -1$

$y'(1) = 2 \rightarrow c_2 + (2c_3 + 3c_4)e^2 = 2$

$\Big\{ c_1 + c_2 = -1$, $c_2 = 2 \Rightarrow c_1 = -3$

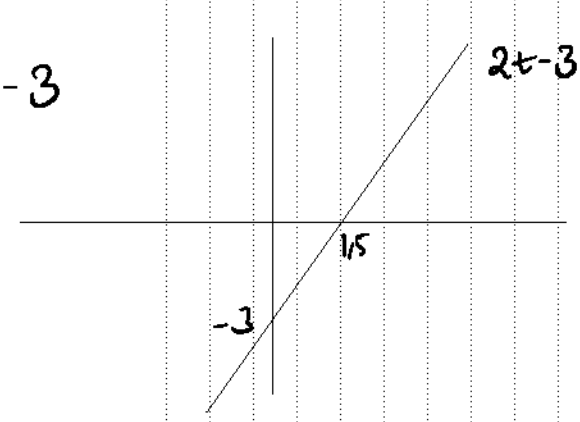
$y''(1) = 0 \rightarrow e^2(4c_3 + 8c_4) = 0$

$y'''(1) = 0 \rightarrow e^2(8c_3 + 20c_4) = 0$

$\Big\{ 2c_3 + 4c_4 = 0$, $2c_3 + 5c_4 = 0$

$c_3 = 0$, $c_4 = 0$

$y = 2t - 3$



$2t-3 \to \infty$ as $t \to \infty$.

4.2. 37. Show that the general solution of $y^{(4)} - y = 0$ can be written as

$$y = c_1 \cos t + c_2 \sin t + c_3 \cosh t + c_4 \sinh t.$$

Determine the solution satisfying the initial conditions $y(0) = 0$, $y'(0) = 0$, $y''(0) = 1$, $y'''(0) = 1$. Why is it convenient to use the solutions $\cosh t$ and $\sinh t$ rather than e^t and e^{-t}?

char eqn $r^4 - 1 = 0$

$(r^2-1)(r^2+1) = 0$

$r_{1,2} = \pm i \to y_1 = \cos t$, $y_2 = \sin t$

$r_{3,4} = \pm 1 \to y_3 = e^t$, $y_4 = e^{-t}$

$\cosh t = \frac{e^t + e^{-t}}{2}$, $\sinh t = \frac{e^t - e^{-t}}{2}$

$y_3 = e^t = \cosh t + \sinh t$, $y_4 = e^{-t} = \cosh t - \sinh t$

gen. soln: $y = c_1 \cos t + c_2 \sin t + c_3 \cosh t + c_4 \sinh t$

IVs: $y(0) = 0 \to c_1 + c_3 = 0$

$y'(0) = 0 \to c_2 + c_4 = 0$

$y''(0) = 1 \to -c_1 + c_3 = 1$

$y'''(0) = 1 \to -c_2 + c_4 = 1$

← one of c_3 or c_4 vanishes at $x=0$

$c_1 = c_2 = -\frac{1}{2}$

$c_3 = c_4 = \frac{1}{2}$ (earlier to derive)

IVP soln: $y = -\frac{1}{2}(\cos t + \sin t) + \frac{1}{2}(\underbrace{\cosh t + \sinh t}_{e^t})$

In each of Problems 1 through 8, determine the general solution of the given differential equation.

4.3. 2. $y^{(4)} - y = 3t + \cos t$

char eqn : $r^4 - 1 = 0$

$(r^2-1)(r^2+1) = 0$

$r_{1,2} = \pm 1$ → $y_1 = \cos t$, $y_2 = \sin t$

$r_{3,4} = \pm 1$ → $y_3 = e^t$, $y_4 = e^{-t}$

$Y(t) = (At+B) + t^1(C\cos t + D\sin t)$ ← the one has multiplicity 1

$Y^{(4)}(t) = 4C\sin t + Ct\cos t - 4D\cos t + Dt\sin t$

$Y^{(4)} - Y = 4C\sin t - 4D\cos t - At - B = 3t + \cos t$

$C = 0$, $D = -\frac{1}{4}$, $A = -3$, $B = 0$

particular soln

$Y(t) = -3t - \frac{1}{4}t\sin t$

In each of Problems 13 through 18, determine a suitable form for $Y(t)$ if the method of undetermined coefficients is to be used. Do not evaluate the constants.

4.3. 13. $y''' - 2y'' + y' = t^2 + 2e^t$

char eqn $r^3 - 2r^2 + r = 0$

$r(r-1)^2 = 0$

$r_1 = 0$ → $y_1 = 1$

$r_{2,3} = 1$ → $y_2 = e^t$, $y_3 = te^t$

16. $y^{(4)} + 4y'' = \sin 2t + t^2 + 4$

char eqn $r^4 + 4r^2 = 0$

$r^2(r^2+4) = 0$

$r_{1,2} = 0$ → $y_1 = 1$, $y_2 = t$

$r_{3,4} = \pm 2i$ → $y_3 = \cos 2t$, $y_4 = \sin 2t$

(check next page)

$$Y(t) = t(At^3 + Bt^2 + Ct + D) + t^2 Ee^t \qquad \text{(soln of 13)}$$

$$Y(t) = At^2 + (Bt + C)e^t + t(D\sin 2t + E\cos 2t)$$

4.4.13. Given that x, x^3, and $1/x$ are solutions of the homogeneous equation corresponding to

$$y''' + \frac{x^2}{x^3}y'' - \frac{2x}{x^3}y' + \frac{2}{x^3}y = \frac{2x^4}{x^3}, \quad x > 0,$$

determine a particular solution.

hom. soln: $y_{hom} = c_1 x + c_2 x^2 + c_3 \frac{1}{x}$

3 unknown u, v, w are fncs of x.

$$Y(t) = \boxed{u y_1 + v y_2 + w y_3}$$

$$Y'(t) = \boxed{u y_1' + v y_2' + w y_3'} + \underbrace{u' y_1 + v' y_2 + w' y_3}_{=0} \quad 1^{st} \text{ constr.}$$

$$Y''(t) = \boxed{u y_1'' + v y_2'' + w y_3''} + \underbrace{u' y_1' + v' y_2' + w' y_3'}_{=0} \quad 2^{nd} \text{ constr.}$$

$$Y'''(t) = \boxed{u y_1''' + v y_2''' + w y_3'''} + \underbrace{u' y_1'' + v' y_2'' + w' y_3''}_{=2x} \quad 3^{rd} \text{ constr. (from DE)}$$

$=0$ when plugged in DE

$$\begin{bmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{bmatrix} \begin{bmatrix} u' \\ v' \\ w' \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 2x \end{bmatrix} = g(x)$$

$$W_1 = \begin{vmatrix} 0 & y_2 & y_3 \\ 0 & y_2' & y_3' \\ 2x & y_2'' & y_3'' \end{vmatrix}$$

$$u' = \frac{W_1}{W}$$

(Cramer's Rule)

$$W = \begin{vmatrix} x & x^2 & \frac{1}{x} \\ 1 & 2x & -\frac{1}{x^2} \\ 0 & 2 & \frac{2}{x^3} \end{vmatrix} = x \begin{vmatrix} 2x & -\frac{1}{x^2} \\ 2 & \frac{2}{x^3} \end{vmatrix} - 1 \begin{vmatrix} x^2 & \frac{1}{x} \\ 2 & \frac{2}{x^3} \end{vmatrix}$$

$$= x\left(\frac{4}{x^2} + \frac{2}{x^2}\right) - 1\left(\frac{2}{x} - \frac{2}{x}\right) = \frac{6}{x}$$

$$W_1 = \begin{vmatrix} 0 & x^2 & \frac{1}{x} \\ 0 & 2x & -\frac{1}{x^2} \\ 2x & 2 & \frac{2}{x^3} \end{vmatrix} = 2x \begin{vmatrix} x^2 & \frac{1}{x} \\ 2x & -\frac{1}{x^2} \end{vmatrix} = 2x(-1-2) = -6x$$

$$W_2 = \begin{vmatrix} x & 0 & \frac{1}{x} \\ 1 & 0 & -\frac{1}{x^2} \\ 0 & 2x & \frac{2}{x^3} \end{vmatrix} = 2x\left[-\left(-\frac{1}{x} - \frac{1}{x}\right)\right] = 4$$

$$W_3 = \begin{vmatrix} x & x^2 & 0 \\ 1 & 2x & 0 \\ 0 & 2 & 2x \end{vmatrix} = 2x\left(2x^2 - x^2\right) = 2x^3$$

$$u' = \frac{W_1}{W} \qquad v' = \frac{W_2}{W} \qquad w' = \frac{W_3}{W}$$

$$= \frac{-6x}{6/x} \qquad = \frac{4}{6/x} \qquad = \frac{2x^3}{6/x}$$

$$= -x^2 \qquad = \frac{2}{3}x \qquad = \frac{x^4}{3}$$

$$u = -\frac{x^3}{3} + c_1 \qquad v = \frac{x^2}{3} + c_2 \qquad w = \frac{x^5}{15} + c_3$$

gen. soln:

$$y(t) = uy_1 + vy_2 + wy_3$$

$$= \underbrace{-\frac{x^4}{3} + \frac{x^4}{3} + \frac{x^4}{15}}_{\text{part soln}} \underbrace{+ c_1 x + c_2 x^2 + c_3 \frac{1}{x}}_{\text{hom. soln.}}$$

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & \cdots & \\ a_{31} & a_{32} & \cdots & \\ a_{41} & a_{42} & \cdots & \end{vmatrix}$$

Taking determinant wrt the second column

$$= -a_{12}\begin{vmatrix} a_{21} & a_{23} & a_{24} \\ a_{31} & a_{33} & a_{34} \\ a_{41} & a_{43} & a_{44} \end{vmatrix} + a_{22}\begin{vmatrix} \cdots \end{vmatrix} - a_{32}\begin{vmatrix} \cdots \end{vmatrix} + a_{42}\begin{vmatrix} \cdots \end{vmatrix}$$

$W(y_1, y_2) = c \cdot e^{\int -p(x)dx}$

5.2. 2 $y'' - xy' - y = 0, \quad x_0 = 0$

(a) Seek power series solutions of the given differential equation about the given point x_0; find the recurrence relation.

(b) Find the first four terms in each of two solutions y_1 and y_2 (unless the series terminates sooner).

(c) By evaluating the Wronskian $W(y_1, y_2)(x_0)$, show that y_1 and y_2 form a fundamental set of solutions.

(d) If possible, find the general term in each solution.

P, Q, R polynomials & $P(x_0) \neq 0 \Rightarrow x_0$ is an ordinary pt.

$$y = \sum_{n=0}^{\infty} a_n x^n \longrightarrow \text{Taylor series expansion of } y \text{ about } x=0$$

$$y = \sum_{n=0}^{\infty} a_n (x-a)^n \longrightarrow \text{expansion about } x = a$$

$$= a_0 + a_1(x-a) + a_2(x-a)^2 + \cdots$$

a) Assume $y = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + a_3 x^3 \cdots$

$$y' = \sum_{n=1}^{\infty} n a_n x^{n-1} = 0 + a_1 + 2a_2 x + \cdots$$

$$y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} = 0 + 0 + 2a_2 + \cdots$$

$$0 = y'' - xy' - y = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} - x\sum_{n=1}^{\infty} n a_n x^{n-1} - \sum_{n=0}^{\infty} a_n x^n$$

$$= \sum_{\substack{n=2 \\ k=0}}^{\infty} \overbrace{n(n-1)}^{k+2} \overbrace{a_n}^{k+2} \overbrace{x^{n-2}}^{k} - \sum_{n=1}^{\infty} n a_n x^n - \sum_{n=0}^{\infty} a_n x^n$$

$k = n-2$

$k+1 = n-1$

$k+2 = n$

$$= \sum_{k=0}^{\infty} (k+2)(k+1)\, a_{k+2}\, x^k - \sum_{n=1}^{\infty} n\, a_n\, x^n - \sum_{n=0}^{\infty} a_n\, x^n$$

$$= \sum_{n=0}^{\infty} (n+2)(n+1)\, a_{n+2}\, x^n - \sum_{n=1}^{\infty} n\, a_n\, x^n - \sum_{n=0}^{\infty} a_n\, x^n$$

$$\left(2 \cdot 1 \cdot a_2 x^0 + \cdots x^1 \cdots\right) - \left(1 a_1 x^1 + \cdots x^2 \cdots\right) - \left(a_0 x^0 + a_1 x^1 \cdots\right)$$

$$= 2a_2 + \sum_{n=1}^{\infty} (n+2)(n+1)\, a_{n+2}\, x^n \quad - - - - \quad - a_0 - \sum_{n=1}^{\infty} a_n x^n$$

$$0 = 2a_2 - a_0 + \sum_{n=1}^{\infty} \left[(n+2)(n+1)\, a_{n+2} - n a_n - a_n\right] x^n$$

$\forall n$ coefe of $x^n = 0 \xrightarrow{①} 2a_2 - a_0 = 0$ (coef of x^0)

$\xrightarrow{②} (n+2)(n+1)\, a_{n+2} - n a_n - a_n = 0$ (coef of x^n)

① $a_2 = a_0/2$

② $(n+2)(n+1)\, a_{n+2} = (n+1)\, a_n$

$a_{n+2} = a_n / n+2$ ⟶ recurrence relation

$a_0, a_1, a_2, a_3, a_4, a_5, a_6, \ldots$

$$y = \sum_{n>0}^{\infty} a_n x^n = \underbrace{\left(a_0 + a_2 x^2 + a_4 x^4 + \cdots\right)}_{k=0,1,2,\ldots} + \underbrace{\left(a_1 x + a_3 x^3 + a_5 x^5 + \cdots\right)}_{k=0,1,2,\ldots}$$

$$= \sum_{k=0}^{\infty} \underbrace{a_{2k}}_{\text{written in } a_0} x^{2k} + \sum_{k=0}^{\infty} \underbrace{a_{2k+1}}_{\text{written in } a_1} x^{2k+1}$$

$a_0 y_0 + a_1 y_1$

(b)

$a_{n+2} = a_n / n+2$

$a_0 = a_0$

$n=0$ $a_2 = a_0/2$

$n=2$ $a_4 = a_2/4 = a_0/4.2$

$n=4$ $a_6 = a_4/6 = a_2/6.4$

$= a_0/6.4.2$

$a_8 = a_0/8.6.4.2$ (4 3 2 1 · 2^4)

$a_{2k} = a_0 / 2k\cdot(2k-2)(2k-4)\cdots 4.2$

$= a_0 / 2^k . k!$

$a_1 = a_1$

$n=1$ $a_3 = a_1/3$

$n=3$ $a_5 = a_3/5 = a_1/5.3$

$a_7 = a_5/7 = a_3/7.5 = a_1/7.5.3$

$a_{2k+1} = a_1 / (2k+1)(2k-1)\cdots 5.3.1$

$= \dfrac{a_1 \quad 2k\cdot(2k-2)\cdots 4.2 \quad = 2^k k!}{(2k+1)\,(2k-1)\cdots 5\,3\,1}$

(2k, 2k-2, 4, 2)

$= a_1 \dfrac{2^k k!}{(2k+1)!}$

$$y = \sum_{n>0}^{\infty} a_n x^n = \sum_{k=0}^{\infty} a_{2k} x^{2k} + \sum_{k=0}^{\infty} a_{2k+1} x^{2k+1}$$

$$= a_0 \sum_{k=0}^{\infty} \frac{1}{2^k . k!} x^{2k} + a_1 \sum_{k>0}^{\infty} \frac{2^k k!}{(2k+1)!} x^{2k+1}$$

(c)

$$W(y_1,y_2)(x)=\begin{vmatrix} \sum_{k=0}^{\infty}\frac{1}{2^k\cdot k!}x^{2k} & \sum_{k=0}^{\infty}\frac{2^k\cdot k!}{(2k+1)!}x^{2k+1} \\ \sum_{k=1}^{\infty}\frac{2k}{2^k\cdot k!}x^{2k-1} & \sum_{k=0}^{\infty}\frac{2^k\cdot k!(2k+1)}{(2k+1)!}x^{2k} \end{vmatrix}$$

$$\left(\sum_{k=0}^{\infty}\frac{1}{2^k\cdot k!}x^{2k}\right)' = \left(\frac{1}{2^0\cdot 0!}x^0+\frac{1}{2^1\cdot 1!}x^2+\dots\right)' = \frac{1\cdot 2}{2^1\cdot 1!}x \quad \checkmark$$

y_1 $\qquad$ y_2

$$W(y_1,y_2)(0)=\begin{vmatrix} \sum_{k=0}^{\infty}\frac{1}{2^k\cdot k!}x^{2k} & \sum_{k=0}^{\infty}\frac{2^k\cdot k!}{(2k+1)!}x^{2k+1} \\ \sum_{k=1}^{\infty}\frac{2k}{2^k\cdot k!}x^{2k-1} & \sum_{k=0}^{\infty}\frac{2^k\cdot k!(2k+1)}{(2k+1)!}x^{2k} \end{vmatrix}_{x=0}$$

y_1' $\qquad$ y_2'

$$= \begin{vmatrix} \frac{1}{2^0\cdot 0!}\,1 & 0 \\ 0 & 1\,\frac{2^0\cdot 0!\,(1)}{1!} \end{vmatrix} = 1$$

$$\sum_{n}^{\infty} a_n x^n = a_0 + a_1x^1+a_2x^2\dots$$

Abel's Thm: $I=\mathbb{R}$

$$W(y_1,y_2)=c\cdot e^{\int x\,dx}=c\cdot e^{\frac{x^2}{2}}$$

$$W(0)=c\cdot \underbrace{e^{\frac{0}{2}}}_{=1}=1 \Rightarrow c=1$$

$$W(t)=e^{x^2/2}$$

5.2 19. (a) By making the change of variable $x - 1 = t$ and assuming that y has a Taylor series in powers of t, find two series solutions of

$$y'' + (x-1)^2 y' + (x^2 - 1)y = 0$$

in powers of $x - 1$.

Any $x \in \mathbb{R}$ is an ordinary pt here!

$t+2 = (x+1)(x-1)\,t$

(b) Show that you obtain the same result by assuming that y has a Taylor series in powers of $x - 1$ and also expressing the coefficient $x^2 - 1$ in powers of $x - 1$.

(a) $x - 1 = t$

$\frac{dx}{dt} = 1$

$\frac{dy}{dt} = \frac{dy}{dx} \cdot \underbrace{\frac{dx}{dt}}_{=1}$

$\frac{d^2y}{dt^2} = \frac{d}{dt}\left(\frac{dy}{dt}\right) = \frac{d}{dt}\left(\frac{dy}{dx}\right)$

$= \frac{d}{dx}\left(\frac{dy}{dx}\right) \cdot \underbrace{\frac{dx}{dt}}_{1}$

DE in t variable:

$\frac{d^2y}{dt^2}$

$$y'' + t^2 y' + (t^2 + 2t)\, y = 0$$

Assume $y = \sum_{n=0}^{\infty} a_n t^n$, $y' = \sum_{n=1}^{\infty} n a_n t^{n-1}$, $y'' = \sum_{n=2}^{\infty} n(n-1) a_n t^{n-2}$

$$\sum_{n=2}^{\infty} n(n-1) a_n t^{n-2} + t^2 \sum_{n=1}^{\infty} n a_n t^{n-1} + (t^2 + 2t) \sum_{n=0}^{\infty} a_n t^n = 0$$

$$\sum_{\substack{n=2 \\ k=n-2}}^{\infty} n(n-1) a_n t^{n-2} + \sum_{\substack{n=1 \\ k=n+1}}^{\infty} n a_n t^{n+1} + \sum_{\substack{n=0 \\ k=n+2}}^{\infty} a_n t^{n+2} + 2 \sum_{\substack{n=0 \\ k=n+1}}^{\infty} a_n t^{n+1} = 0$$

$$\sum_{k=0}^{\infty} (k+2)(k+1)\, a_{k+2} t^k + \sum_{k=2}^{\infty} (k-1)\, a_{k-1} t^k + \sum_{k=2}^{\infty} a_{k-2} t^k + 2 \sum_{k=1}^{\infty} a_{k-1} t^k = 0$$

$$2 \cdot 1 \cdot a_2 t^0 + 3 \cdot 2 \cdot a_3 t^1 + 2 a_0 t^1 + \sum_{k=2}^{\infty} \left[(k+2)(k+1)\, a_{k+2} + (k-1)\, a_{k-1} + a_{k-2} + 2 a_{k-1} \right] t^k = 0$$

t^0 terms: $2a_2 = 0$

t^1 terms: $6a_3 + 2a_0 = 0$

t^k terms: $(k+2)(k+1)a_{k+2} + (k-1)a_{k-1} + a_{k-2} + 2a_{k-1} = 0$

$k \geq 2$

$$(k+2)(k+1)a_{k+2} + (k+1)a_{k-1} + a_{k-2} = 0$$

$$a_{k+2} = -\frac{a_{k-1}}{(k+2)} - \frac{a_{k-2}}{(k+2)(k+1)}$$

$a_0, a_1, a_2 = 0, a_3 = -\frac{a_0}{3}$

$k=2$ $\quad a_4 = \frac{-3a_1 - a_0}{4 \cdot 3} = -\frac{a_1}{4} - \frac{a_0}{4.3}$

$k=3$ $\quad a_5 = \frac{-4a_2 - a_1}{5.4} = -\frac{a_2}{5} - \frac{a_1}{5.4}$

$$a_6 = \frac{-5a_3 - a_2}{6.5} = -\frac{a_3}{6} - \frac{a_2}{6.5} = \frac{a_0}{19}$$

$\frac{5}{(3+2)a_0}$ / $7.3.6.2$

$$a_7 = \frac{-6a_4 - a_3}{7.6} = -\frac{a_4}{7} - \frac{a_3}{7.6} = \frac{a_1}{7.4} + \frac{a_0}{7.4.3} + \frac{a_0}{7.6.3}$$

$$a_8 = \frac{-7a_5 - a_4}{8.7} = -\frac{a_5}{8} - \frac{a_4}{8.7} = \frac{a_1}{8.5.4} + \frac{a_1}{8.7.4} + \frac{a_0}{8.7.4}$$

$$y = a_0 t^0 + a_1 t^1 + a_2 t^2 + a_3 t^3 + \cdots + a_8 t^8 + \cdots$$

$$= a_0 t^0 + a_1 t^1 - \frac{a_0}{3} t^3 - \left(\frac{a_1}{4} + \frac{a_0}{12}\right) t^4 - \left(\frac{a_1}{20}\right) t^5 + \frac{a_0}{18} t^6$$

$$+ \left(\frac{a_1}{28} + \frac{5a_0}{252}\right) t^7 + \cdots$$

$$= a_0\left(1 - \frac{1}{3}t^3 - \frac{1}{12}t^4 + \frac{1}{18}t^6 + \frac{5}{252}t^7 + \cdots\right)$$
$$+ a_1\left(t - \frac{1}{4}t^4 - \frac{1}{20}t^5 + \frac{1}{28}t^7 \cdots\right)$$

(b) $y = \sum_{n=0}^{\infty} a_n (x-1)^n$

$y' = \sum_{n=1}^{\infty} n a_n (x-1)^{n-1}$

$y'' = \sum_{n=2}^{\infty} n(n-1) a_n (x-1)^{n-2}$

$$y'' + (x-1)^2 y' + (x^2-1)y = 0$$

$$\sum_{n=2}^{\infty} n(n-1) a_n (x-1)^{n-2} + (x-1)^2 \sum_{n=1}^{\infty} n a_n (x-1)^{n-1} + (x^2-1)\sum_{n=0}^{\infty} a_n (x-1)^n = 0$$

$(x-1)^{n+1}$

??

$$x^2 - 1 = (x-1)^2 + 2(x-1)$$
$$t^2 \quad + 2t$$

exactly the same thing

Answer: Plug in $t = x-1$ in part (a)

READ THM 5.3.1 FIRST!!

In each of Problems 1 through 4, determine $\phi''(x_0)$, $\phi'''(x_0)$, and $\phi^{(4)}(x_0)$ for the given point x_0 if $y = \phi(x)$ is a solution of the given initial value problem.

5.3. 2. $y'' + (\sin x)y' + (\cos x)y = 0; \quad y(0) = 0, \quad y'(0) = 1$

$$y''(0) + \underbrace{\sin 0}_{0} \cdot \underbrace{y'(0)}_{1} + \underbrace{\cos 0}_{1} \cdot \underbrace{y(0)}_{0} = 0 \quad \Rightarrow \quad y''(0) = 0$$

$$y''' + \underbrace{(\cos x)}_{1}\underbrace{y'}_{1} + \underbrace{(\sin x)}_{0}\underbrace{y''}_{0} - \underbrace{(\sin x)}_{0}\underbrace{y}_{0} + \underbrace{(\cos x)}_{1}\underbrace{y'}_{1} = 0$$

$$y'''(0) = -2, \quad \text{similarly } y^{(4)}(0) = 0. \ (\text{ex})$$

In each of Problems 5 through 8, determine a lower bound for the radius of convergence of series solutions about each given point x_0 for the given differential equation.

5.3. 5. $y'' + 4y' + 6xy = 0; \quad x_0 = 0, \quad x_0 = 4$

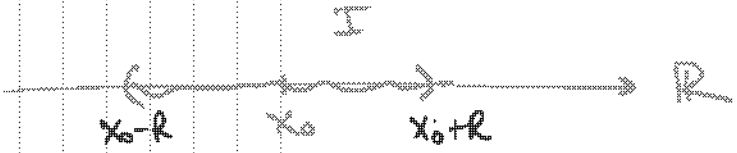
$\frac{Q(x)}{P(x)} = 4$ $\quad \frac{R(x)}{P(x)} = 6x = 6(x-4) + 24$

both are analytic everywhere. $R = \infty$

analytic at x_0 : $\exists I \quad x_0 \in I$ and

- f Taylor expandable on I (about x_0) $\Leftrightarrow$ the Taylor series converges on I
- f has a Taylor series expansion on I

How to find I?



ROOT TEST
RATIO TEST
(to find the radius of conv R for power series)

you should check for convergence at these points separately.

Also check Legendre Eqn.

10. The Chebyshev Equation. The Chebyshev differential equation is

$$\underbrace{(1-x^2)}_{P(x)}y'' \underbrace{-xy'}_{Q(x)} \underbrace{+\alpha^2 y}_{R(x)} = 0, \qquad (*)$$

where α is a constant.

(a) Determine two solutions in powers of x for $|x| < 1$, and show that they form a fundamental set of solutions.

(b) Show that if α is a nonnegative integer n, then there is a polynomial solution of degree n. These polynomials, when properly normalized, are called the Chebyshev polynomials. They are very useful in problems that require a polynomial approximation to a function defined on $-1 \le x \le 1$.

(c) Find a polynomial solution for each of the cases $\alpha = n = 0, 1, 2, 3$.

$$y'' - \underbrace{\frac{x}{1-x^2}}_{p(x)} y' + \underbrace{\frac{\alpha^2}{1-x^2}}_{q(x)} y = 0$$

by Thm 5.3.1 gen. soln at $x_0 = 0$, here is in form $y = \sum a_n (x-x_0)^n$

⇒ both analytic on $|x|<1$, because:

$$\frac{1}{1-x^2} = \frac{1}{1-(x^2)} = \sum_{m=0}^{\infty} (x^2)^m = \sum_{m=0}^{\infty} x^{2m} \quad \to \text{ conv. for } |x^2|<1 \text{ i.e. for } |x|<1$$

$$\Rightarrow p(x) = -\frac{x}{1-x^2} = -x \sum_{m=0}^{\infty} x^{2m} = -\sum_{m=0}^{\infty} x^{2m+1} \quad \text{and} \quad q(x) = \alpha^2 \sum_{m=0}^{\infty} x^{2m}$$

are Taylor series of p & q in $|x|<1$ region.

(OR) P, Q, R are polynomials. $P \neq 0$ if $|x|<1$ so $\frac{Q}{P}, \frac{R}{P}$ are analytic there.

(a) $y = \sum_{n=0}^{\infty} a_n x^n \qquad y' = \sum_{n=1}^{\infty} n a_n x^{n-1} \qquad y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}$

$$(1-x^2)\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} - x\sum_{n=1}^{\infty} n a_n x^{n-1} + \alpha^2 \sum_{n=0}^{\infty} a_n x^n = 0$$

$$\sum_{\substack{n=2 \\ k=0}}^{\infty} \overbrace{n(n-1)}^{k+2} a_n x^{\overbrace{n-2}^{=k}} - \sum_{n=2}^{\infty} n(n-1) a_n x^n - \sum_{n=1}^{\infty} n a_n x^n + \alpha^2 \sum_{n=0}^{\infty} a_n x^n = 0$$

($n(n-1)$: $k+1$; a_n: $k+2$)

$$\sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n - \sum_{n=2}^{\infty} n(n-1) a_n x^n - \sum_{n=1}^{\infty} n a_n x^n + \alpha^2 \sum_{n=0}^{\infty} a_n x^n = 0$$

$$2a_2 + 6a_3 x - a_1 x + \alpha^2 (a_0 + a_1 x)$$

$$+ \sum_{n=2}^{\infty} (n+2)(n+1) a_{n+2} x^n - \sum_{n=2}^{\infty} n(n-1) a_n x^n - \sum_{n=2}^{\infty} n a_n x^n + \alpha^2 \sum_{n=2}^{\infty} a_n x^n = 0$$

$$\sum_{n=2}^{\infty} \left[(n+2)(n+1) a_{n+2} - n(n-1) a_n - n a_n + \alpha^2 a_n \right] x^n$$

$$2a_2 + \alpha^2 a_0 = 0 \longrightarrow a_2 = -\frac{\alpha^2}{2} a_0$$

$$6a_3 - a_1 + \alpha^2 a_1 = 0 \longrightarrow a_3 = \frac{1-\alpha^2}{6} a_1$$

$$\forall n \geq 2, \quad (n+2)(n+1) a_{n+2} + (\alpha^2 - n^2) a_n = 0$$

$$a_{n+2} = \frac{n^2 - \alpha^2}{(n+2)(n+1)} a_n$$

$n=2$:
$$a_4 = \frac{4-\alpha^2}{4 \cdot 3} a_2 = -\frac{(4-\alpha^2)\alpha^2}{4 \cdot 3 \cdot 2} a_0$$

$n=4$:
$$a_6 = \frac{16-\alpha^2}{6 \cdot 5} a_4 = -\frac{(16-\alpha^2)(4-\alpha^2)\alpha^2}{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2} a_0$$

$$y = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + \cdots$$

$$= a_0 + a_1 x - \frac{\alpha^2}{2} a_0 x^2 + \frac{1-\alpha^2}{6} a_1 x^3 - \frac{(4-\alpha^2)\alpha^2}{4 \cdot 3 \cdot 2} a_0 x^4 + \cdots$$

$$= a_0 \left(1 - \frac{\alpha^2}{2} x^2 - \frac{(4-\alpha^2)\alpha^2}{4 \cdot 3 \cdot 2} x^4 + \cdots + \frac{\left((n-2)^2 - \alpha^2\right)\left((n-4)^2 - \alpha^2\right) \cdots \alpha^2}{n!} x^n + \cdots \right)$$

($n=0$, $n=2$, $n=4$, $n=2m$, $m \in \mathbb{Z}$; even)

$$\frac{1}{n!} \prod_{k=1}^{n/2} \left[(n-2k)^2 - \alpha^2 \right]$$

$$+ a_1 \left(x + \frac{1-\alpha^2}{6} x^3 + \frac{(9-\alpha^2)(1-\alpha^2)}{5!} + \cdots + \frac{\left((n-2)^2 - \alpha^2\right) \cdots (1-\alpha^2)}{n!} x^n + \cdots \right)$$

($n=1$, $n=3$, $n=5$, $n=2m+1$, $m \in \mathbb{Z}$; odd)

$$\alpha^2 \left(2^2 - \alpha^2\right)\left(4^2 - \alpha^2\right)\left(6^2 - \alpha^2\right) \cdots \left(10^2 - \alpha^2\right)\left(12^2 - \alpha^2\right)$$

$\alpha = k \in \mathbb{Z}$

$$y_0 = 1 - \frac{\alpha^2}{2}x^2 - \frac{(4-\alpha^2)\alpha^2}{4.3.2}x^4 + \cdots + \frac{((2m-2)^2-\alpha^2)((2m-4)^2-\alpha^2)\cdots\alpha^2}{(2m)!}x^{2m} + \cdots$$

$$y_1 = x + \frac{1-\alpha^2}{6}x^3 + \frac{(1-\alpha)(3^2-\alpha^2)}{5!}x^5 + \cdots + \frac{((2m-1)^2-\alpha^2)\cdots(1-\alpha^2)}{(2m+1)!}x^{2m+1} + \cdots$$

fund. set of solns on $(-1,1)$?? $\quad W(y_0,y_1)(0) = \begin{vmatrix} y_0(0) & y_1(0) \\ y_0'(0) & y_1'(0) \end{vmatrix}$

$y_0(0) = 1$ $\qquad y_0' = 0 - \alpha^2 x - \frac{(4-\alpha^2)\alpha^2}{3.2}x^3 + \cdots$

$y_1(0) = 0$ $\qquad y_0'(0) = 0$

$y_1' = 1 + \frac{1-\alpha^2}{2}x^2 + \cdots$

$y_1'(0) = 1$

$$W(y_0,y_1)(0) = \begin{vmatrix} y_0(0) & y_1(0) \\ y_0'(0) & y_1'(0) \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1 \neq 0 \quad \checkmark$$

(b) $\alpha \in \mathbb{Z} \Rightarrow$ If α = even, then y_0 terminates at $x^{|\alpha|}$

α = odd, then y_1 terminates at $x^{|\alpha|}$

(c) $\alpha = 0 \longrightarrow y_0 = 1$

$\alpha = 1 \longrightarrow y_1 = x$

$\alpha = 2 \longrightarrow y_0 = 1 - 2x^2$

$\alpha = 3 \longrightarrow y_1 = x - \frac{4}{3}x^3$